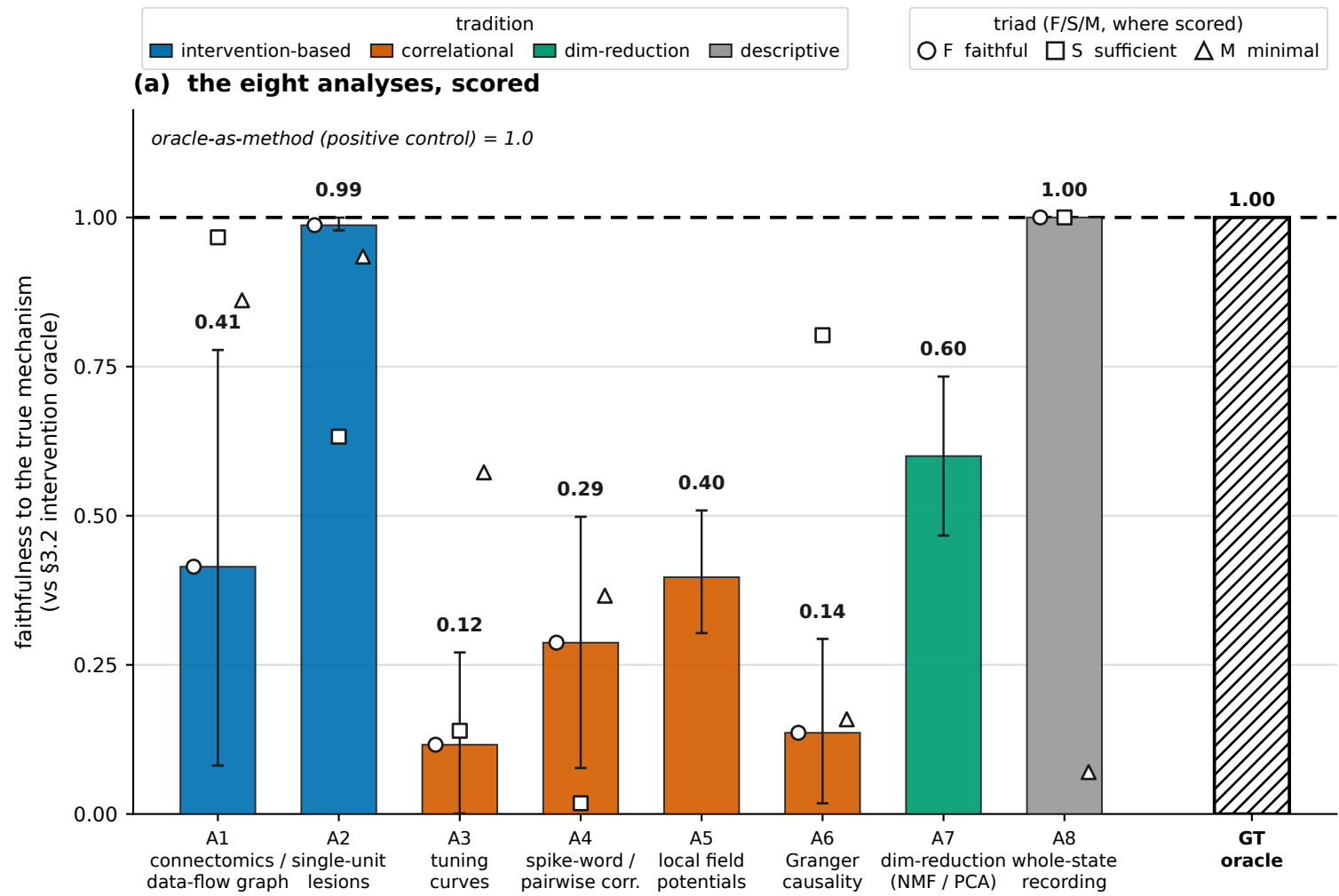


The neuroscience battery on a known machine — scored against ground truth

Jonas & Kording's battery, quantified: classical analyses find rich structure yet score low in faithfulness to the true mechanism.



Data: committed \$R records (6 core games); whiskers = 95% bootstrap CI. See Supplement provenance table.

(b) key findings

A6 Granger $F = 0$

Pong: false-edge rate = 1.00,
missed-edge = 1.00 $\rightarrow$ triad $F = 0.00$.

Temporal precedence $\neq$ causation. on a

A3 spurious tuning

clock-locked machine Granger infers
spurious bidirectional edges everywhere.
Pong spurious-tuning rate = 1.00;
Space Invaders = 0.67, Q*bert = 0.53.
Strongly-tuned units whose tuning does

A2 $p = 0.99$, blind to interactions

NMF match their true causal role —
the classic tuning-curve trap, quantified.
Single-unit lesion importance $p = 0.99$
vs the true role — yet single lesions
miss 40% of interactions (Space Inv.).

A7 super-additive pair Δ up to 1044
NMF > PCA (both mediocre)

invisible to one-unit-at-a-time ablation.
NMF matched-component fraction = 0.60
vs PCA = 0.60 mean (NMF best 0.8).
Non-negativity aligns components with
the additive register basis better than